

Practice Questions from Week 2

Question 3 : 640653738034

 View Solutions (0)

Total Mark : 4.00 | Type : MCQ

Let $k : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be a valid kernel. Is $\sin(x_1) \cos(x_2)k(x_1, x_2)$ a valid kernel? Here, $x_1, x_2 \in \mathbb{R}$?



OPTIONS :

☐ Yes

☐ No

Solution (3)

Given $k(x_1, x_2)$ is a valid kernel.

Multiplying the function $\sin(x_1)\cos(x_2)$ with $k(x_1, x_2)$ is not valid function as $\sin(x_1)\cos(x_2) k(x_1, x_2) \neq \sin(x_2)\cos(x_1) k(x_2, x_1)$ so symmetricity fails.

Question 4 : 640653738035

[View Solutions \(0\)](#)

Total Mark : 4.00 | Type : MCQ

Consider the following kernel:

$$k : R^2 \times R^2 \rightarrow R$$
$$k(x, y) = (x^T y)^2 + 1$$

Which of the following transformation mapping ϕ may correspond to the kernel k ?

OPTIONS :

- ☐ $\phi([x_1, x_2]^T) = [x_1^2, \sqrt{2}x_1x_2, x_2^2, 1]^T$
- ☐ $\phi([x_1, x_2]^T) = [x_1^2, x_1 + x_2, x_2^2, 1]^T$
- ☐ $\phi([x_1, x_2]^T) = [x_1, \sqrt{2}x_1^2x_2^2, x_2, 1]^T$
- ☐ $\phi([x_1, x_2]^T) = [x_1, x_1x_2, x_2, 1]^T$

Solution (4)

Take any two arbitrary vector in $\mathbb{R}^2$ $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$

apply kernel transformation $k(x, y)$

$$\begin{aligned}
 &= \left(\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \right)^2 + 1 \\
 &= (x_1 y_1 + x_2 y_2)^2 + 1 \\
 &= x_1^2 y_1^2 + 2x_1 y_1 x_2 y_2 + x_2^2 y_2^2 + 1 \\
 &= \begin{bmatrix} x_1^2 & \sqrt{2}x_1 x_2 & x_2^2 & 1 \end{bmatrix} \begin{bmatrix} y_1^2 \\ \sqrt{2}y_1 y_2 \\ y_2^2 \\ 1 \end{bmatrix} \\
 &= \phi(x)^T \phi(y)
 \end{aligned}$$

Question 5 : 640653738036

 **View Solutions (0)** Total Mark : 4.00 | Type : MCQ

Consider a dataset consisting of 1000 samples in a 20-dimensional space. You apply Kernel PCA with a polynomial kernel of degree 3 to reduce the dimensionality of the data. Which of the following statements regarding Kernel PCA is true?

OPTIONS :

- ☐ Kernel PCA always results in a lower-dimensional representation of the data compared to standard PCA.
- ☐ Kernel PCA can only be applied to datasets that are linearly separable.
- ☐ Choosing a higher degree polynomial kernel in Kernel PCA can lead to increased flexibility in capturing non-linear relationships.
- ☐ None of these.

Solution (5)

- Kernel PCA does not always result in a lower-dimensional representation than standard PCA as it depends on the number of principal components chosen.
- Kernel PCA is not limited to linearly separable datasets. In fact, its strength is in handling non-linear structures.
- Choosing a higher degree polynomial kernel in Kernel PCA can lead to increased flexibility in capturing non-linear relationships. Kernel PCA is an extension of PCA that uses kernel methods to project data into a higher-dimensional space, where linear PCA is then performed. Using a higher-degree polynomial kernel allows Kernel PCA to capture more complex non-linear relationships in the data.

Question 3 : 640653827434[View Solutions \(0\)](#) Total Mark : 3.00 | Type : MCQ

Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ be two vectors in $\mathbb{R}^2$. Let k_1 and k_2 be two functions defined from $\mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$:

$$k_1(\mathbf{x}, \mathbf{y}) = x_1y_1 + x_2y_2 + x_1x_2y_1y_2$$

$$k_2(\mathbf{x}, \mathbf{y}) = x_1y_1 + x_2y_2 + x_1x_2y_1y_2 + 1$$

Which of the following statements is true?

OPTIONS :

- ☐ Both k_1 and k_2 are valid kernels.
- ☐ k_1 is a valid kernel, but k_2 is not a valid kernel.
- ☐ k_2 is a valid kernel, but k_1 is not a valid kernel.
- ☐ Neither k_1 nor k_2 is a valid kernel.

Solution (3)

Approach :

To verify validity of any kernel, express $k(x, y)$ as a dot product in some feature space.

$$\text{Given, } x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \text{ and } y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \in \mathbb{R}^2$$

$$k_1(x, y) = x_1 y_1 + x_2 y_2 + x_1 x_2 y_1 y_2$$

$$k_1(x, y) = (x_1, x_2, x_1 x_2) \times (y_1, y_2, y_1 y_2) \quad [\text{observe, they are transformed into 3d from 2d}]$$

$$k_1(x, y) = \phi(x)^T \phi(y)$$

$\therefore k_1(x, y)$ is a valid kernel

$$k_2(x, y) = x_1 y_1 + x_2 y_2 + x_1 x_2 y_1 y_2 + 1$$

$$k_2(x, y) = (x_1, x_2, x_1 x_2, 1) \times (y_1, y_2, y_1 y_2, 1) [\text{observe, they are transformed into 4d from 2d}]$$

$$k_2(x, y) = \phi(x)^T \phi(y)$$

$\therefore k_2(x, y)$ is a valid kernel

Question 5 : 640653827436[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : SA

Kernel PCA with a polynomial kernel of degree 3 is run on a dataset of 100 data-points. Let ϕ be the transformation corresponding to this kernel. If the largest eigenvalue of the centered kernel matrix is 10, find the largest eigenvalue of the covariance matrix of the transformed dataset, where ϕ is used to effect the transformation.

Answer (Numeric):

Solution (5)

we know,

$$\text{eigen value of } \frac{1}{n} \phi(x) \phi(x)^T = \lambda$$

$$\text{eigen value of } \phi(x)^T \phi(x) = 10$$

and the fact that non – zero eigen values of both the matrix are same

$$\implies n\lambda = 10$$

$$\implies 100\lambda = 10$$

$$\implies \lambda = \frac{10}{100} = 0.1$$

Question 7 : 640653827437

View Solutions (0)

Total Mark : 4.00 | Type : SA

$k : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ is a polynomial kernel of degree 4. Find the value of $k\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}\right)$.



Answer (Numeric):

Answer

Solution (7)

$$k(x, y) = (x^T y + 1)^4$$

$$\Rightarrow k\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}\right) = \left(\begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 1\right)^4 = (2 - 1 + 1)^4 = 2^4 = 16$$

Question 2 : 640653992018

[View Solutions \(0\)](#) Total Mark : 4.00 | Type : MCQ

Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ be vectors in $\mathbb{R}^2$. Define the functions k_1 and k_2 as:

$$k_1(\mathbf{x}, \mathbf{y}) = x_1 y_1 + x_2 y_2 + (x_1 + x_2)(y_1 + y_2)$$

$$k_2(\mathbf{x}, \mathbf{y}) = x_1 y_1 + x_2 y_2 + (x_1^2 + y_2^2) + 3$$

Which of the following statements is true?

OPTIONS :

- ☐ Both k_1 and k_2 are valid kernels.
- ☐ k_1 is a valid kernel, but k_2 is not a valid kernel.
- ☐ k_2 is a valid kernel, but k_1 is not a valid kernel.
- ☐ Neither k_1 nor k_2 is a valid kernel.

Solution(2)

[Link to the discourse post](#)

Method 1 - applying explicit phi transformation



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Oct 2024

We have:

$$k_1(x, y) = x_1 y_1 + x_2 y_2 + (x_1 + x_2)(y_1 + y_2)$$

Consider $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ as follows:

$$\phi(z_1, z_2) = \begin{bmatrix} z_1 \\ z_2 \\ z_1 + z_2 \end{bmatrix}$$

Then:

$$\begin{aligned} \phi(x_1, x_2)^T \phi(y_1, y_2) &= \begin{bmatrix} x_1 & x_2 & x_1 + x_2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_1 + y_2 \end{bmatrix} \\ &= x_1 y_1 + x_2 y_2 + (x_1 + x_2)(y_1 + y_2) \\ &= k_1(x, y) \end{aligned}$$

Method 2 - Mercers Theorem

Condition 1 – K is symmetric i. e; $K(x, y) = K(y, x)$

$$k_2(x, y) = x_1 y_1 + x_2 y_2 + (x_1^2 + y_2^2) + 3$$

$$k_2(y, x) = y_1 x_1 + y_2 x_2 + (y_1^2 + x_2^2) + 3$$

$$\because (x_1^2 + y_2^2) \neq (y_1^2 + x_2^2)$$

$\therefore k_2(x, y)$ fails to hold symmetricity

Since the symmetry condition (1) is already violated, we don't need to check positive semi – definiteness further.

A kernel must be symmetric to be valid, and this kernel fails on that front

Question 3 : 640653992020

[View Solutions \(0\)](#) Total Mark : 4.00 | Type : MCQ

Consider the following kernel:

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$$k(x, y) = (x^T y)^2 + 1$$

Which of the following transformation mapping ϕ may correspond to the kernel k ?

OPTIONS :

- ☐ $\phi([x_1, x_2]^T) = [x_1^2, \sqrt{2}x_1x_2, x_2^2, 1]^T$
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- ☐ $\phi([x_1, x_2]^T) = [x_1, \sqrt{2}x_1^2x_2^2, x_2, 1]^T$
- ☐ $\phi([x_1, x_2]^T) = [x_1, x_1x_2, x_2, 1]^T$

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apply kernel transformation $k(x, y)$

$$= \left(\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \right)^2 + 1$$

$$= (x_1 y_1 + x_2 y_2)^2 + 1$$

$$= x_1^2 y_1^2 + 2x_1 y_1 + x_2^2 y_2^2 + 1$$

$$= \begin{bmatrix} x_1^2 & \sqrt{2}x_1x_2 & x_2^2 & 1 \end{bmatrix} \begin{bmatrix} y_1^2 \\ \sqrt{2}y_1y_2 \\ y_2^2 \\ 1 \end{bmatrix}$$

$$= \phi(x)^T \phi(y)$$

Question 4 : 640653992019

 **View Solutions (0)** Total Mark : 3.00 | Type : MCQ

Given n data points in a d -dimensional space with a non-linear relationship, we apply kernel PCA to reduce the dimensionality and select the first k principal components. Is it possible for k to be greater than d ?



OPTIONS :

☐ yes

☐ No

Solution(4)

- Yes, in kernel PCA, it is possible for k to be greater than d . This is because kernel PCA works by mapping the data into a higher-dimensional feature space through the kernel trick. In this transformed space, the number of principal components (k) can actually exceed the original input dimension (d).
- The kernel function implicitly maps the data to a higher (potentially infinite) dimensional space where linear PCA is then performed, allowing for more principal components than the original dimensions.